# AUSTRALIAN INSTITUTE
OF HIGHER EDUCATION

**Unit Code:** MBIS5004

**Unit Name:** Cyber Intelligence

**Study Period:** B7 **Year** 2024

# Unit Outline

## Subject Information

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|---|---|
| **Term Dates** | Monday 14 October - Friday 08 November |
| **Session Times** | Tue/Fri 8:30 AM SYD |
| **Awards** | Master of Business Information Systems (Cyber Security) |
| **Prerequisites/ Corequisites** | Nil |
| **Delivery Method** | Face-to-face / Blended / Online |
| **Unit Description** | MBIS5005 Cyber Intelligence introduces the students to a comprehensive training and in-depth study in the field of digital/computer forensics. The unit combines training on both theoretical and practical aspects with up-to-date forensics software tools. The students will get the opportunity to build a firm foundation in this field and at the same time, having excellent hands-on experience that is needed in the current world. The student will learn various techniques and methods used in recovering and producing evidence from electronic devices.<br><br>The subject starts by introducing the students with the operation systems, file types and data structures. The students will learn data acquisition process, techniques to process evidence, digital forensics analysis and validation process, E-mail and social media investigations, electronic evidence collection process from mobile and computer equipment and so on. The other key areas of this unit include introduction to digital forensics and understanding investigation, contemporary digital forensic issues and trends, forensics process, and forensic validation of result, processing crime and incident scenes, E-discovery, guidelines, and standards, report writing for high-tech investigations, expert testimony and ethics, physical, environmental, and organisational considerations for deploying forensics computing initiatives, mobile device forensics and the Internet of Anything, Virtual Machine forensics, live acquisitions, and network forensics, challenges inhibiting a successful digital forensic investigation etc. This unit will contribute to prepare for professional certification as forensic examiner GCFE (https://www.giac.org/digitalbadges ). |

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**Subject Information (cont'd)**

| Learning Outcomes and Graduate Attributes | On successful completion, students will have achieved the following learning outcomes: | |
|---|---|---|
| | **Unit Learning Outcomes** | **AIH Graduate Attributes** |
| | ULO01: Outline a comprehensive process of digital forensics theory and practice and document evidence and report on computer forensics findings. | 1. Communication: The ability to communicate persuasively, both orally and in writing, with a diverse range of audiences.<br><br>2. Collaboration: The ability to build and manage teams, and to liaise and co-operate with stakeholders successfully. |
| | ULO02: Explore the functionalities and challenges of various types of file system related to digital forensics. | 3. Research: The ability to conduct and evaluate project.<br><br>4. Critical thinking & problem solving: The ability to pro-actively identify and solve problems creatively and in a structured and methodical way. |
| | ULO03: Analyse and demonstrate the current ethical and legal aspects, forensics ethical behavior and professional conduct. | 5. Ethical behaviour: The ability to pro-actively identify and respond to ethical issues and to set standards of ethical behaviour.<br><br>6. Flexibility: The ability to critically assess, evaluate and synthesise alternatives. |
| | ULO04: Administer appropriate tools and techniques to carry out a digital forensics' investigations including recovery, validation and presentation of evidence. | 7. Leadership: The ability to lead others effectively. |
| | ULO05: Perform a practical investigation into computer forensics, including industry and legal procedures with respect to data acquisitions. | 8. Self-sustained learning: The ability to conduct future self-sustained learning through accurate and advanced research. |

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| **Contact Details** | **Unit Co-ordinator** | Dr. Mohammed Al-Zobbi (m.al-zobbi@aih.edu.au) |
| | **Lecturer & Tutor** | Dr. Mohammed Al-Zobbi (m.al-zobbi@aih.edu.au) |
| **Resources** | **Prescribed Textbooks** | Nelson, B, Phillips, A & Steuart, C 2018, Guide to Computer Forensics and Investigations, 6th Edn, Course Technology Cengage Learning.<br><br>Note: prescribed textbook must be available in the library or through the library e-book service. |
| | **Reference Textbooks** | Sachowski, J 2019, Implementing Digital Forensic Readiness: From Reactive to Proactive Process, 2nd Edn, CRC Press.<br><br>Whitman, ME & Matord, HJ 2016, Management of Information Security, 5th Edn, Cengage Learning.<br><br>Vacca, JR 2017, Computer and Information Security Handbook, 3rd Edn, Morgan Kaufmann.<br><br>Smith, R 2015, Elementary Information Security, 2nd Edn, Jones & Barlett Learning.<br><br>Note: recommended textbook must be available in the library or through the library e-book service. |
| | **Reference Journals** | Szewczyk, P, Sansurooah, K & Williams, P 2018, ‘ An Australian Longitudinal Study Into Remnant Data Recovered From Second-Hand Memory Cards ’, International Journal of Information Security and Privacy (IJISP), vol. 12, no.4, pp. 82 – 97, doi:10.4018/IJISP.2018100106.<br><br>Shortall, A, & Azhar, MA 2015, ‘ Forensic Acquisitions of WhatsApp Data on Popular Mobile Platforms ’, In: IEEE. Sixth International Conference on Emerging Security Technologies (EST), Braunschweig, Germany, IEEE, pp. 13 – 17.<br><br>Kolhe, M & Ahirao, P 2017, ‘ Live Vs Dead Computer Forensic Image Acquisition ’, International Journal of Computer Science and Information Technologies, vol. 8, no. 3, https://ijcsit.com/docs/Volume%208/vol8issue3/ijcsit2017080331.pdf |
| | **Recommended websites:** | ACPO. 2015, ‘ Good Practice Guide for Computer-Based Electronic Evidence ’, Available from: https://www.7safe.com/docs/default-source/default-document-library/acpo_guidelines_computer_evidence_v4_web.pdf [Accessed 30th April 2021]<br><br>LSoft Technologies Inc.(n.d.). NTFS Basics. Available from: https://www.ntfs.com/ntfs_basics.htm [Accessed 28th April 2021]<br><br>LSoft Technologies Inc.(n.d.). FAT File System. Available from: http://www.ntfs.com/fat-systems.htm [Accessed 28th April 2021]<br><br>NIST. (2019, November 15). Computer Forensics Tool Testing Program (CFTT). NIST; NIST. Available from: https://www.nist.gov/itl/ssd/software-quality-group/computer-forensics-tool-testing-program-cftt [Accessed 28th April 2021] |

The AIH Referencing Guide is available to download. Go to the Library and Learning Support Moodle page.

## Weekly Workshop Schedule

| Week | Week Starting | Session Reading Materials | Workshop Activities | Assessment Task Due |
|---|---|---|---|---|
| 1 | Session 1 | Introduction to Digital Forensics<br><br>Planning and Report Writing for Forensics Investigation | Group discussion and Q/A | |
| | Session 2 | Data Acquisition | Group discussion and Q/A | A1 - Lab 1 |
| 2 | Session 3 | File Systems in Forensics | Group discussion and Q/A | |
| | Session 4 | Digital Forensics Analysis<br><br>Virtual Machines, Network Forensics and Live Acquisition | Group discussion and Q/A | A1 - Lab 2 |
| 3 | Session 5 | Windows Forensics<br><br>Data Recovery | Group discussion and Q/A | |
| | Session 6 | Mobile Forensics<br><br>Ethics in forensics and Ethical challenges | Group discussion and Q/A | A2 - Quiz 1<br>A1 - Lab 3 |
| 4 | Session 7 | Email & Internet Forensics | Work on group report submission | A3 - Report and presentation |
| | Session 8 | Review | Feedback on group report | A2 - Quiz 2 |
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## Assessment Details

| Assessment Number & Type | Assessment Description | Week/ Session Assessed & Due Date | Weighting | Learning Outcomes Assessed | Submission Requirements |
|---|---|---|---|---|---|
| A1<br>Portfolio<br>Individual | Students will build up a portfolio based on their weekly understanding of concepts. Students will read one application case study and or applied a short model from their prescribed book/e-text and write a short recommendation for decision makers (150 words). Responses will be discussed in class creating an engaging environment. | Sessions 2, 3, 4, 6, 7 | 35% | ULO01<br>ULO02<br>ULO04 | Moodle<br>150 words<br>Turnitin |
| A2<br>Quiz | Quiz 1 assessing concepts up to session 4.<br><br>Quiz 2 assessing concepts in sessions 5-7. | Sessions 5 and 8 | 30% | ULO01<br>ULO02<br>ULO03<br>ULO04<br>ULO05 | In Class<br>20 minutes |
| A3<br>Report<br>Group Work | Analyse and report a cyber intelligence case study by evaluating at least three different cyber intelligence techniques in a given business case. | Session 7<br>5 PM | 25% | ULO01<br>ULO03<br>ULO04<br>ULO05 | Moodle<br>3000 words<br>Turnitin |
| A3<br>Participation<br>Group Work | Group presentation based on Report (A3). | Session 7<br>5 PM | 10% | ULO01<br>ULO02<br>ULO03<br>ULO04<br>ULO05 | Moodle<br>20 minutes<br>Turnitin |

**NOTE: Pass requirements**

**Students must make a reasonable attempt in their assessment events in order to be eligible to pass the subject and must achieve a combined result from these assessment events of at least 50% to pass the subject.**

**If a student fails any individual piece of assessment they will be eligible to continue in the subject and attain a pass in the subject, provided their combined result from all assessment events in the subject is 50% or above.**

**You can view details on your assessments at Moodle in the Assessment Brief and in-class instruction.**

## Overall Grade Assessment

| Grade | Marking Criteria/Standards |
|---|---|
| **High Distinction**<br>(outstanding performance)<br>Code: HD<br>Mark range: 85% and above | Complete and comprehensive understanding of the unit content; development of relevant skills to an outstanding level; demonstration of an extremely high level of interpretive and analytical ability and intellectual initiative; and excellent achievement of all major and minor objectives of the unit. |
| **Distinction**<br>(very high level of performance)<br>Code: D<br>Mark range: 75-84% | Very high level of understanding of the unit content; development of relevant skills to a very high level; demonstration of a very high level of interpretive and analytical ability and intellectual initiative; and comprehensive achievement of all major and minor objectives of the unit. |
| **Credit**<br>(high level of performance)<br>Code: C<br>Mark range: 65-74% | High level of understanding of the unit content; development of relevant skills to a high level; demonstration of a high level of interpretive and analytical ability and achievement of all major objectives of the unit; some minor objectives not fully achieved. |
| **Pass**<br>(competent level of performance)<br>Code: P<br>Mark range: 50-64% | Adequate understanding of most of the basic unit content; development of relevant skills to a satisfactory level; adequate interpretive and analytical ability and achievement of all major objectives of the unit; some minor objectives not achieved. |
| **Non-graded Pass**<br>Code: NGP | Successful completion of a unit assessed on a pass/fail basis, indicating satisfactory understanding of unit content; satisfactory development of relevant skills; satisfactory interpretive and analytical ability and achievement in all major objectives of the unit. |
| **Fail - unsatisfactory performance**<br>Code: F<br>Mark range: below 50% | Inadequate understanding of the basic unit content; failure to develop relevant skills; insufficient evidence of interpretive and analytical ability; and failure to achieve some or all major and minor objectives of the unit. |
| **Fail - no assessment submitted**<br>Code: FNS | Did not present any work for assessment, to be considered as failed. |
| **Grade Pending**<br>Code: GP | A final grade is yet to be awarded for the unit as students have not yet completed all the assessment tasks and special consideration has been approved due to medical grounds or exceptional circumstances beyond the control. This is a temporary grade only and must be finalised before the end of the following semester. |
| **Supplementary Exam**<br>Code: SX | A final grade is yet to be awarded for the unit as a supplementary exam has been approved for students who have failed the unit and obtained total mark between (46-49%), provided the student has completed all other assessment units. This is a temporary grade only and must be finalised before the end of the following semester. |
| **Withdraw with Failure**<br>Code: WF | Cancelled enrolment in the unit after the final date for withdrawal without failure.<br>Cancelled enrolment in the unit due to non-payment of fees. |
| **Withdraw Without Failure**<br>Code: AW | Cancelled enrolment in the unit before the final date for withdrawal without failure. This grade may also be awarded to students who withdraw from a unit after the withdrawal date under special or compassionate circumstances. In these cases, the grade is awarded at the discretion of the Teaching and Learning Committee.<br>A unit with the grade of AW does not appear on a student's academic transcript. |
| **Advanced Standing**<br>Code: AS | Credit has been granted for the unit of study following an application and its approval for Advanced Standing |

**Policies & Forms**

All students should be familiar with the relevant policies and how they apply to their study. Policies and forms can be accessed on the Australian Institute of Higher Education website at: www.aih.nsw.edu.au

### ACCESS AND EQUITY

The Australian Institute of Higher Education is committed to providing equal opportunity and promoting inclusive practices and processes for all students and clients within the limits of its resources.

### SPECIAL CONSIDERATION

You may seek Special Consideration in the evaluation of your subject assessments where you feel such matters as serious illness, or other exceptional circumstances beyond your control that may have affected your performance. A Special Consideration form must be submitted within 3 days of the assessment due date or event. Please contact Student Services (studentservices@aih.nsw.edu.au) for a copy of the application form (https://aih.nsw.edu.au/about-us/policies-procedures/).

### ATTENDANCE POLICY

The Australian Institute of Higher Education emphasizes the need for all students to attend classes on a regular and consistent basis to develop the skills and attitudes necessary to compete in the highly competitive labor market. Students who are not in attendance for at least 50 percent of any scheduled class will be considered absent for that class. Lateness is a disruption of a good learning environment and is discouraged.

### ACADEMIC INTEGRITY

The Australian Institute of Higher Education considers academic honesty to be one of its highest values. Students are expected to be the sole authors of their work (except in the case of group assignments). Use of another person's work or ideas must be accompanied by specific citations and references. The Australian Institute of Higher Education subscribes to a third-party plagiarism detection service, and reserves the right to check all student work to verify that it meets the guidelines of this policy. Academic dishonesty is a serious offense and may result in a range of penalties. Procedures for processing plagiarism, contract cheating and exam cheating offenses are described in the Academic Misconduct Policy and Procedure (https://aih.nsw.edu.au/about-us/policies-procedures/).

### ACADEMIC APPEALS

The Student Grievance Handling and Resolution Policy and Procedure policy outlines the process for a fair, equitable and confidential means of resolving complaints and appealing academic decisions (https://aih.nsw.edu.au/about-us/policies-procedures/).

### REVIEW OF ASSESSMENT

Where a student believes their assessment instrument has been marked incorrectly, they may apply to have that assessment instrument re-marked once only. Requests must be made within five (5) days of results being released. The Student Assessment Policy and Procedure provides instructions on requesting re-marks (https://aih.nsw.edu.au/about-us/policies-procedures/).

### LATE POLICY

An assessment item submitted after the assessment due date, without an approved extension or without approved mitigating circumstance, will suffer a late submission penalty. The late submission penalty is a reduction of the mark allocated for the assessment item by 5% per day (or part thereof) of the total marks available for the assessment item as indicated in the Unit Outline. **A "day" for this purpose is defined as any day of the week including weekends**. Assignments submitted later than one (1) week after the due date will not be accepted (https://aih.nsw.edu.au/about-us/policies-procedures/).